

SCIENCE SCHOLARS ACADEMY

DIFFERENTIATION OF EXPONENTIAL & LOGARITHMIC FUNCTIONS:

Lesson targets:

- (1). Sketching exponential & logarithmic functions. ✓✓
- (2). $\frac{d}{dx}$ (exponential functions) ✓✓
- (3). $\frac{d}{dx}$ (logarithmic functions) ✓✓
- (4). $\frac{d}{dx} (a^{f(x)}) \rightarrow$ A form of exponential functions. ✓✓
- (5). Mixed & More complex cases. ✓✓
 $y = \ln e^x$ ✓✓ $y = \ln \sin^{-1} x$ ✓✓
- (6). Research & Assignment ✓✓
- (7). Plan for first principles ✓
- (8). Next stop & END

SCIENCE SCHOLARS ACADEMY

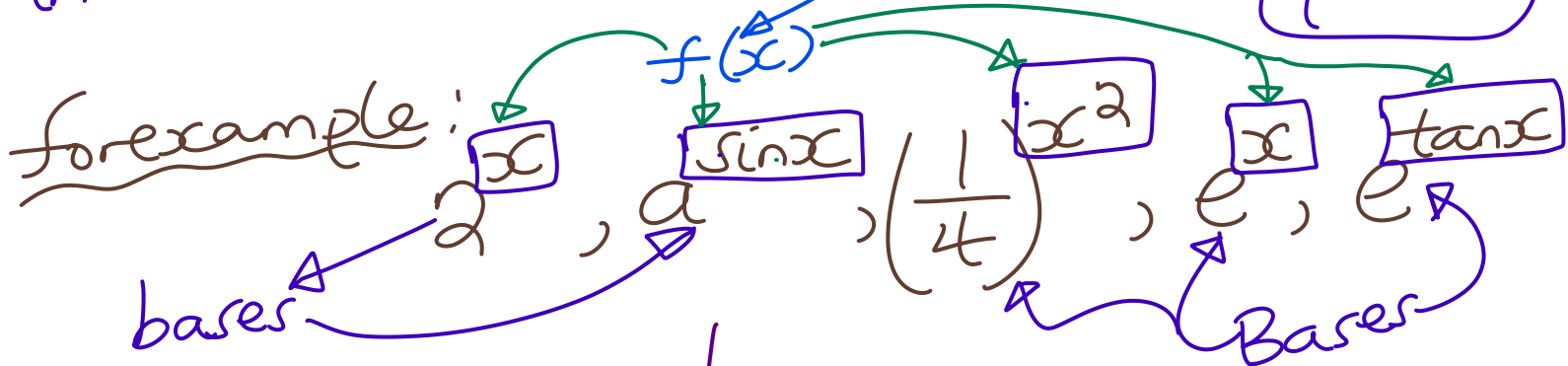
Plan for first principles

✓ Algebraic >> Trig >> Inverse Trig >>
Exponential >> Logarithmic { L I A T E }

EXPONENTIAL FUNCTIONS.

→ Recall; Exponent can mean @ power in math;

→ So; Exponential functions are those in which the function is a power.



Note: for this section, we usually concentrate on functions where the base is e such as e^x , e^{x^2} , $e^{\sin x}$, etc

The rest i.e 2^x, 3^x, 4 ^{$\sin x$} , 5 ^{$\sqrt{x}$} are handled later, ✓

Scalar bases. ✓

SCIENCE SCHOLARS ACADEMY

Note: The usual general format of exponential functions is: (For now),

$$e^{f(x)}$$

base ✓

function having x ✓

Qn: Sketch a graph of e^x , e^{2x} or $e^{\sin x}$.

Advice:

→ Get any values of x for e^x ; get $x=0, 1, 2, 3, \dots$
 → for trig e.g. $e^{\sin x}$ → $5^\circ, 10^\circ, 15^\circ, \dots$

→ Tabulate

$e^x \rightarrow \{y = e^x\}$

x	0	1	2	3
e^x	1.0	2.7	7.4	20.1

$e^{\sin x} \{y = e^{\sin x}\}$

x	0°	15°	30°	45°
$\sin x$	0.0	0.3	0.5	0.7
$e^{\sin x}$	1.0	1.3	1.6	2.0

Shift

+

ln

→

e^x

Means Power

So you just add the power.

Alpha

+

ln

→

e

Needs a power;

So;

Press that e^{Δ} {Put the power}

Hence

SCIENCE SCHOLARS ACADEMY

How do we press e^2

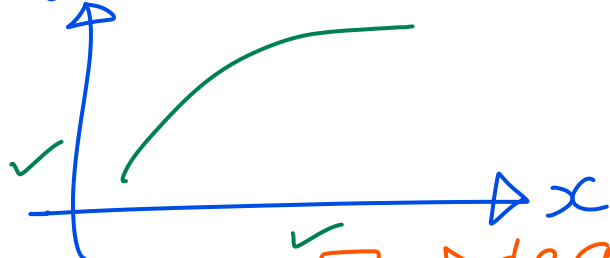
①. Shift + $\ln$ + 2 = e^2 ✓ ✓

②. Alpha + $\ln$ + 1 + 2 = e^2 ✓ ✓

To plot

$$y = e^x$$

Sketch ①.



if $x = 30$



degrees

{ CALC. Must be in DEG }

$\sin x$

if $x = 1$ { no unit } → { degrees }

(4 times)

CALC → { Rad }

Note; Press

MODE

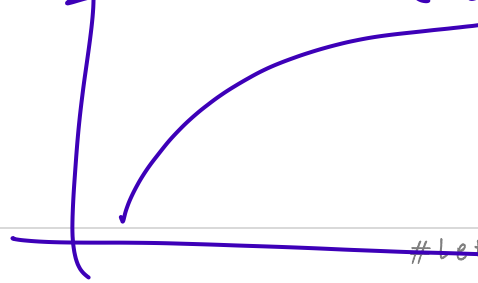
until you find

DEG (1)

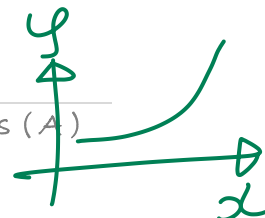
RAD (2)

$$y = e^{\sin x}$$

Sketch ②



✓



SCIENCE SCHOLARS ACADEMY

$\frac{d}{dx}$ (Exponential functions)

Rule No. 1;

$$\frac{d}{dx}(e^x) = e^x$$

→ why? → Evidence from first principles. ✓

Recall; → Long-cut of chain-Rule. ✓
→ Using it, find $\frac{dy}{dx}$ for $y = e^x$.

$y = e^x$ → Let $u = x$ ✓

$y = e^u$ → $\frac{dy}{du} = e^u$

$\frac{du}{dx} = 1$

used & dumped. ✓

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = e^u (1) = e^u$$

$$\frac{dy}{dx} = e^x \quad \checkmark$$

SCIENCE SCHOLARS ACADEMY

(2). $y = e^{2x}$. find $\frac{dy}{dx}$.

$y = e^{2x}$ → $u = 2x$
 $\frac{du}{dx} = 2$ ✓

$y = e^u \rightarrow \frac{dy}{du} = e^u$ ✓

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = e^u(2) = 2e^u$ ✓

$\frac{dy}{dx} = 2e^{2x}$ ✓

(3). $y = e^{x^2}$ ✓. find $\frac{dy}{dx}$.

$y = e^{x^2}$ → Let; $u = x^2$

$y = e^u \rightarrow \frac{dy}{du} = e^u$ ✓

$\frac{du}{dx} = 2x$ ✓

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = e^u(2x) = 2xe^u$ ✓

$\frac{dy}{dx} = 2xe^{x^2}$ ✓

Qn $\rightarrow$ Find dy/dx

SCIENCE SCHOLARS ACADEMY

(4). $y = e^{\sin x}$ $\rightarrow$ Let $u = \sin x$

$y = e^u \rightarrow \frac{dy}{du} = e^u \checkmark$

$\frac{du}{dx} = \cos x \checkmark$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = e^u \cos x = \boxed{e^{\sin x} \cos x} \checkmark \checkmark$$

COMPARE :

$y = f(x)$ vs. $\frac{dy}{dx}$ OR $f'(x)$

$y = e^x \checkmark$

vs.

$\frac{dy}{dx} = e^x \checkmark$

$y = e^{2x} \checkmark$

vs.

$\frac{dy}{dx} = 2e^{2x} \checkmark$

$y = e^{x^2} \checkmark$

vs.

$\frac{dy}{dx} = 2x e^{x^2} \checkmark$

$y = e^{\sin x} \checkmark$

vs.

$\frac{dy}{dx} = \cos x e^{\sin x} \checkmark$

$y = e^{\tan x} \checkmark$

vs.

$\frac{dy}{dx} = \sec^2 x e^{\tan x} \checkmark$

SCIENCE SCHOLARS ACADEMY

(*) $y = e^{\sin^2 x} \rightarrow \frac{dy}{dx} = \sin 2x e^{\sin^2 x}$ ✓✓

$\frac{d}{dx} [(\sin x)^2] = 2(\sin x)^1 \cos x$
 $= 2 \sin x \cos x$
 $\sin 2x$

(*) $y = e^{\tan^{-1} x} \rightarrow \frac{dy}{dx} = \left(\frac{1}{1+x^2} \right) e^{\tan^{-1} x}$ ✓✓

Assgn $\rightarrow$ find $\frac{dy}{dx}$.

(a). $y = e^{x^4}$

(b). $y = e^{5x}$

(c). $y = e^{\left(\frac{2}{x}\right)}$

(d). $y = e^{-x^{-3/4}}$

(e). $y = e^{\cos^3 x}$

(f). $y = e^{\sin^{-1} x}$

(g). $y = e^{\tan 4x}$

(h). $y = e^{\cot 3x}$

(i). $y = e^{\sec \frac{1}{3} x}$

SCIENCE SCHOLARS ACADEMY

LOGARITHMIC ✓✓

FUNCTIONS: ✓✓

(Local categories)

Artificial Logarithms ✓
{ Have any base except }
 e
e.g $\log_5 3$, $\log_y x$, $\log_a 4$
etc ✓

Natural Logarithms ✓
{ Have base }
 e
 $\log_e 3$ ($\log_e y$)
Base ✓

Note: $\log_e x$

Log to the base e can be replaced with $\ln$ ✓✓ $\ln$

Small L $\rightarrow$ n $\rightarrow$ natural
Logarithm ✓

Read as; $\ln(e \text{ l e n})$ ✓

50

SCIENCE SCHOLARS ACADEMY

Uganda

$\log_e 3 \rightarrow \ln$

$\ln 3$ ✓

Read as $\underline{\text{Lin}}$ of 3
Natural log of 3 ✓

* $\ln x^2 = \log_e x^2$

* $\log_e \sin x = \ln \sin x$

Sketch (a) $\ln x$ MEANS $y = \ln x$ ✓

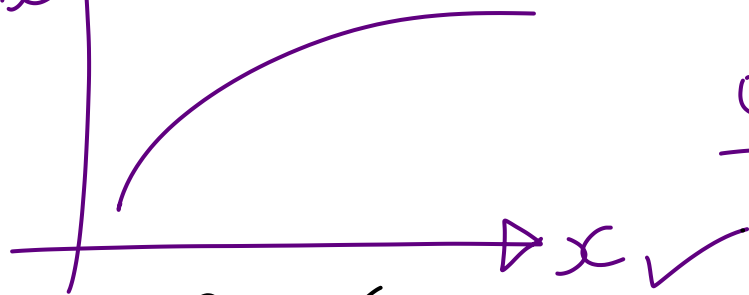
x	0	1	2	3
y = $\ln x$	—	0	0.7	1.1

(b) Sketch $y = \ln \cos 4x$ ✓

x	✓	✓	✓	✓
4x				
$\cos 4x$				
$\ln \cos 4x$				

SCIENCE SCHOLARS ACADEMY

✓
 $y = \ln x$



OR



$\frac{d}{dx}$ (Logarithmic functions)

(a). Natural Log $\{ \ln \}$ ✓

General format of a natural logarithmic function is;

$$\ln[f(x)]$$

Natural
Log

of

→ the function

$\sin x$ ✓

#etssolvesciences(A)

$\sin^{-1} x$ ✓

e^x ✓

SCIENCE SCHOLARS ACADEMY

RULE no. 1

Learn it by heart for now!

$$\frac{d}{dx}(\ln x) = \frac{1}{x}$$

why? → first principles!

BRINGING IN THE LONG WT.

(a). $y = \ln x$ → $u = x$

$$\frac{du}{dx} = 1$$

$y = \ln u$ → $\frac{dy}{du} = \frac{1}{u}$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = \frac{1}{u} (1) = \frac{1}{u} \rightarrow x$$

$$\frac{dy}{dx} = \frac{1}{x} \quad \checkmark$$

SCIENCE SCHOLARS ACADEMY

b. $y = \ln(2x)$ $\rightarrow u = 2x$ $\frac{1}{2x}$
 $\frac{du}{dx} = 2$ ✓
 $y = \ln u \rightarrow \frac{dy}{du} = \frac{1}{u}$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = \frac{1}{u}(2) = \frac{2}{u} \rightarrow 2x$$

$$= \frac{2}{2x}$$

$$\left\{ \frac{dy}{dx} = \frac{1}{x} \right\}$$

Hint: If $\ln$ is a natural log;
 then; $\ln a^b = b \ln a$ ✓

Shorter Route ✓

$x \ln y = \ln y^x$ ✓

(3). $y = \ln x^2$

$y = 2 \ln x$ ✓ $\rightarrow y = \ln(x^2)$

$$\frac{dy}{dx} = 2 \left(\frac{1}{x} \right) = \frac{2}{x}$$

$y = \ln u$
 $\frac{dy}{du} = \frac{1}{u}$

$u = x^2$
 $\frac{du}{dx} = 2x$ ✓

$$\frac{dy}{dx} = \frac{1}{u}(2x) = \frac{2x}{x^2} = \frac{2}{x}$$

SCIENCE SCHOLARS ACADEMY

(4). $y = \ln \cos x$ $\rightarrow u = \cos x$

$y = \ln u \rightarrow \frac{dy}{du} = \frac{1}{u}$ $\frac{du}{dx} = -\sin x$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = \frac{1}{u} (-\sin x) = -\frac{\sin x}{\cos x}$$

$$\frac{dy}{dx} = -\tan x$$

(5). $y = \ln 5x^4$

$y = \ln u$
 $\frac{dy}{du} = \frac{1}{u}$ $u = 5x^4$
 $\frac{du}{dx} = 20x^3$

$$\frac{dy}{dx} = \frac{1}{u} (20x^3) = \frac{20x^3}{5x^4}$$

$$\frac{dy}{dx} = \frac{4}{x}$$

(6). $y = \ln \sin 2x$

$u = \sin 2x$
 $\frac{du}{dx} = 2 \cos 2x$

$y = \ln u$
 $\frac{dy}{du} = \frac{1}{u}$

$$\frac{dy}{dx} = \frac{1}{u} (2 \cos 2x) = \frac{2 \cos 2x}{\sin 2x}$$

$$\frac{dy}{dx} = 2 \cot 2x$$

COMPARE

SCIENCE SCHOLARS ACADEMY

$y = f(x)$ rs $\frac{dy}{dx} / f'(x) \checkmark$

$y = \ln x$ rs $\frac{dy}{dx} = \frac{1}{x}$

$y = \ln 2x$ rs $\frac{dy}{dx} = \frac{2}{2x} = \frac{1}{x}$

$y = \ln x^2$ rs $\frac{dy}{dx} = \frac{2x}{x^2} = \frac{2}{x}$

$y = \ln \cos x$ rs $\frac{dy}{dx} = \frac{-\sin x}{\cos x} = -\tan x$

$y = \ln 5x^4$ rs $\frac{dy}{dx} = \frac{20x^3}{5x^4} = \frac{4}{x}$

$y = \ln \sin 2x$ rs $\frac{dy}{dx} = \frac{2 \cos 2x}{\sin 2x} = 2 \cot 2x$

$y = \ln x^8 \rightarrow \frac{dy}{dx} = \frac{8x^7}{x^8} = \frac{8}{x} \checkmark$

$y = \ln \tan x \rightarrow \frac{dy}{dx} = \frac{\sec^2 x}{\tan x}$

$y = \ln e^x \rightarrow \frac{dy}{dx} = \frac{e^x}{e^x} = 1 \checkmark$

BASIC RULE:

SCIENCE SCHOLARS ACADEMY

$$\frac{d}{dx} \left\{ \ln[f(x)] \right\} = \frac{f'(x)}{f(x)}$$

Brain teasers:

- (1). $y = \ln(\sqrt{x})$ Justus $\rightarrow \frac{dy}{dx} = \frac{\frac{1}{2}x^{-\frac{1}{2}}}{\sqrt{x}}$ ✓
- (2). $y = \ln(e^{\sin x})$ Alfred $\rightarrow \frac{dy}{dx} = \frac{\cos x \cdot e^{\sin x}}{e^{\sin x}}$ ✓
- (3). $y = \ln(x^5 - 4x^3)$ Josh $\rightarrow \frac{dy}{dx} = \frac{5x^4 - 12x^2}{x^5 - 4x^3}$
- (4). $y = \ln \cot^2 x$ Judith $\rightarrow \frac{dy}{dx} = \checkmark \checkmark$
- (5). $y = \ln \cos^{-1} x$ $\rightarrow \frac{dy}{dx} = \frac{-1/\sqrt{1-x^2}}{\cos^{-1} x}$ ✓ ✓
- (6). $y = \ln 3$ $\rightarrow \frac{dy}{dx} = 0$ (why) $\ln 3$ is a value on a calculator

TASK: Simplify the answers above individually. ✓

#letsolvesciences(A) constant

SCIENCE SCHOLARS ACADEMY

Cheat code;

$$y = \ln e \rightarrow \log_e e = 1 \checkmark$$

Recall;

$$\log_a a = 1 \checkmark$$

$$\underline{\ln e = 1} \checkmark$$

(*) So;

$$y = \ln e^{\sin x}$$

$$y = \sin x \ln e \quad (1)$$

$$y = \sin x \checkmark \checkmark$$

$$\frac{dy}{dx} = \cos x \checkmark \checkmark$$

$$y = \ln \cot^2 x \rightarrow (\cot x)^2$$

$$\text{derivative} = 2(\cot x)'(-\operatorname{cosec}^2 x)$$

(*) ✓

$$(4) \frac{dy}{dx} = \frac{-2 \cot x \operatorname{cosec}^2 x}{\cot^2 x} \checkmark \checkmark$$

Hint; $\ln(\cot^2 x) = 2 \ln \cot x = 2 \frac{-\operatorname{cosec}^2 x}{\cot x}$

SCIENCE SCHOLARS ACADEMY

Assignment: find dy/dx

(1). $y = \ln e^{\sin^2 x}$

(4). $y = 2 \ln \sin^{-1} x$

(2). $y = \ln \csc^3 x$

(5). $y = \ln \left(\frac{1}{5} \right)$

(3). $y = \ln(8x^4)$

(6). $y = \ln x^{-6}$

TASK → first review Basic Rules, then Long cuts then short cuts

→ DO all numbers the tutor did.

→ Then attack brain teasers & assignments;

RESEARCH: find dy/dx for;

(a). $y = \log_5 x$

(c). $y = \log_3 2x$

(b). $y = 2^x$

(d). $y = 5^{2x}$

SCIENCE SCHOLARS ACADEMY

Next stop → Maclaurin's theory
↓
Integration ^{OR.} 2

OR

Both!

END

Thank you
Paul Tutor

© Science Scholars Academy
Ug

0753279592
(WhatsApp Calls).
#letssolvesciences

SCIENCE SCHOLARS ACADEMY